

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: HealthTech Pulse

Challenge ID: PS-04

Title: Smart Triage & Clinical Decision Support System (CDSS)

1. Executive Summary & Problem Overview

Emergency departments (EDs), out-patient clinics, and rural primary healthcare centers (PHCs) constantly struggle with acute patient overcrowding, high clinician-to-patient ratios, and severe operational latency during intake. Traditional paper-based or manual triage methods frequently lead to inaccurate acuity prioritization, delayed interventions for deteriorating patients, and dangerous clinical diagnostic blind spots.

Furthermore, medical duty officers and nursing staff lack real-time bedside clinical intelligence to rapidly flag adverse drug-drug interactions, detect early signs of sepsis or shock, and summarize extensive longitudinal patient health histories.

The objective of **PS-04 (Smart Triage & Clinical Decision Support System)** is to architect an explainable, privacy-preserving, AI-assisted clinical triage and decision support engine. The platform must rapidly ingest unstructured patient chief complaints, multimodal vital telemetry, and lab metrics to compute standardized triage acuity scores (e.g., ESI / Manchester Triage), generate ranked differential diagnosis possibilities, highlight critical clinical alerts, and seamlessly streamline hospital bed allocation workflows.

2. Core Personas & Access Roles

1. **Triage Nurse / Intake Paramedic:** Ingests patient vital signs, records acute chief complaints via text or ambient voice, and receives instantaneous algorithmic acuity stratification (Level 1–5).
2. **Attending Emergency Physician:** Reviews AI-generated differential diagnosis summaries, examines explainable evidence citations, orders diagnostic labs, and validates suggested therapeutic pathways.
3. **Hospital Operations / Bed Allocator:** Monitors real-time ED queue status, departmental load heatmaps, ICU/critical care bed utilization, and patient transfer pipelines.
4. **Patient / Attendant (Kiosk / Mobile View):** Submits digital self-intake triage symptoms, tracks live queue estimates, and accesses simplified vernacular discharge instructions.

3. Key Functional Modules & Technical Deliverables

Module 1: Multimodal Intake & Standardized Triage Scoring

- **Multimodal Data Capture:** Ingests chief complaints via multi-lingual voice/audio transcription, vernacular free-text entry, and digital vital telemetry (HR, SpO2, NIBP, Temperature, Respiratory Rate, GCS).
- **Automated Acuity Stratification:** Computes standard Emergency Severity Index (ESI Levels 1–5) or Manchester Triage System scores within seconds of intake.
- **Early Deterioration & Sepsis Flagging:** Automatic calculation of validated physiological deterioration indexes (e.g., NEWS2 / qSOFA) with immediate red-alert escalation for unstable patients.

Module 2: Explainable Clinical Decision Support (CDSS) Engine

- **Ranked Differential Diagnosis:** Synthesizes patient demographics, presenting symptoms, and vitals to propose ranked differential diagnostic possibilities.
- **Explainable Evidence Citations:** Provides clinician-verifiable diagnostic rationales linked to standard medical literature (e.g., ICD-10 codes, medical guideline citations) avoiding hallucinated outputs.
- **Drug-Drug & Allergy Contraindication Guardrails:** Bedside scanning mechanism detecting potential medication collisions, adverse contraindications, and known patient allergies.

Module 3: Clinical Notes Summarization & EHR Interoperability

- **Automated SOAP Note Generation:** Transforms unstructured patient-doctor consultations into structured clinical documentation (Subjective, Objective, Assessment, Plan).
- **FHIR / HL7 Interoperability:** Clean JSON export and ingestion conformant with modern healthcare data exchange protocols (HL7 / FHIR standards) and national digital health frameworks (e.g., ABDM).
- **De-identification & HIPAA/PDP Compliance:** On-device or API-level masking of Personally Identifiable Information (PII) before external model inference.

Module 4: Departmental Queue & Hospital Resource Logistics

- **Real-Time ED Flow Dashboard:** Live Kanban/matrix view of patient severity cohorts, average turnaround latency, and pending diagnostic investigations.
- **Predictive Bed & ICU Allocation:** Algorithmic forecasting of patient admission probabilities versus discharge velocity to optimize bed turnover.
- **Emergency Escalation Alerts:** Automated SMS/WhatsApp or PA notification dispatch to on-call specialty consultants when trauma or cardiac resuscitation criteria are triggered.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web / Tablet, React.js (Vite) / Next.js / Tailwind CSS / Lucide Icons

Mobile / Kiosk App, "React Native, Flutter, or Responsive Web PWA"

Backend & APIs, Python (FastAPI / Django) or Node.js (Express) with REST / WebSockets

Database & Caching, PostgreSQL / Supabase / MongoDB + Redis for live queue caching

Clinical NLP & AI Models, "BioBERT / Med-PaLM / Gemini API (1.5 Flash or Pro) with structured JSON schema outputs, LangChain, Whisper API (Voice-to-Text)"

Healthcare Standards, FHIR Python SDK / HAPI FHIR / OpenEMR integration schema

Security & Privacy, "Role-Based Access Control (RBAC), AES-256 Data Encryption, PII

Redaction tools (Presidio)"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Clinical Triage Accuracy & Safety Guardrails (25 Points):** Correctness of acuity stratification (ESI/NEWS2), adherence to medical safety thresholds, and reliable contraindication alerts.
2. **Explainability & Transparency of CDSS (25 Points):** Quality of AI diagnostic reasoning, justification transparency, and verifiable citation back-linking rather than black-box claims.
3. **Workflow Integration & Multi-Persona Usability (20 Points):** Smooth handoff between triage nurse, treating physician, and queue operations; sub-5-second intake efficiency.
4. **Data Privacy, FHIR Standards & Security (15 Points):** Robust handling of patient health data, anonymization/PII masking, and standards-compliant schema design.
5. **Pitch Quality, 24h Sprint Prototype & Code (15 Points):** Functional end-to-end prototype demonstration, clean Git commit history, and comprehensive README documentation.

6. Submission Guidelines

- **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup guide, mock patient datasets, and .env.example.
- **Live Deployment:** Hosted interactive medical portal or web dashboard (Vercel / Render / AWS).
- **Demo Video / Deck:** A crisp 3-minute video showing patient intake, automated ESI acuity assignment, clinical diagnosis recommendation, and FHIR export.